

Project Documentation

Automated Network Request Management in ServiceNow

Team ID/SID : SWUID2026136950

1. Introduction

Project Title:

Automated Network Request Management in ServiceNow

Team Members:

Name	Role
Kharizma A	Developer, Flow Designer, Security Analyst, Tester, Documenter

2. Project Overview

Purpose:

The purpose of this project is to automate the entire network request lifecycle in ServiceNow — from submission to approval, task fulfillment, closure, and feedback collection. The system eliminates manual coordination, provides a full audit trail, and gives all stakeholders real-time visibility.

Features:

1. Dynamic UI Policies for smart form visibility
2. Auto-Create Fulfillment Task on approval
3. 8 Email Notifications across the lifecycle
4. SLA Definition with 4-hour approval tracking
5. Catalog UI Policies for dynamic behavior
6. Request Cancellation
7. SLA Breach Notification
8. Approval Reminder
9. Auto-Escalation for overdue approvals
10. Request Priority Levels
11. Three Reports
12. Management Dashboard
13. Feedback Survey

3. Architecture

Frontend:

Service Portal (HTML5, CSS3, JavaScript, AngularJS) — user-facing catalog item form and request tracking.

Backend:

ServiceNow Platform:

- Flow Designer (workflow automation)
- Business Rules (server-side logic)
- Client Scripts (form behavior)
- ACLs (record-level security)

Database:

ServiceNow Cloud Database (MySQL-based):

- Custom Table: x_2128291_networ_0_network_request
- sc_req_item (Requested Items)
- sc_task (Catalog Tasks)
- sysapproval_approver (Approvals)
- sys_user, sys_user_group (Users and Groups)
- CMDB (Device reference)

4. Setup Instructions

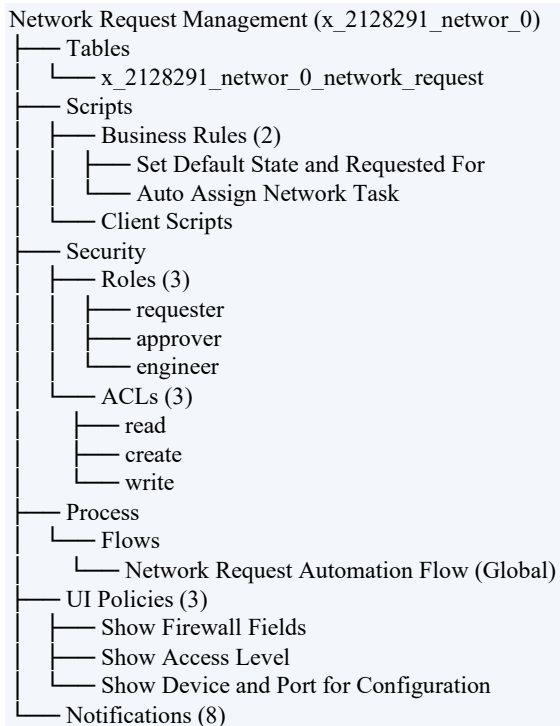
Prerequisites:

1. ServiceNow Developer Instance (PDI)
2. Admin access to the instance
3. Scoped Application: Network Request Management

Installation:

1. Log in to ServiceNow instance as admin
2. Navigate to System Update Sets > Local Update Sets
3. Click New and import the XML file
4. Or upload the Update Set XML directly via Retrieved Update Sets
5. Preview and commit the update set

5. Folder Structure

Application Scope:

6. Running the Application

Steps:

1. Navigate to Service Portal: dev181300.service-now.com/sp
2. Search for "Network Request"
3. Open the catalog item and fill the form
4. Submit the request
5. The flow triggers automatically
6. Approver receives an email
7. Approver approves via My Approvals
8. Task is auto-created for Network Team
9. Engineer closes the task
10. Requested Item auto-closes

7. API Documentation

ServiceNow REST APIs used:

Endpoint	Method	Purpose
/api/now/table/x_2128291_network_0_network_request	GET	Fetch network requests
/api/now/table/sc_req_item	GET	Fetch requested items
/api/now/table/sc_task	GET	Fetch catalog tasks
/api/now/table/sysapproval_approver	GET	Fetch approvals
/api/now/table/cmdm_ci_netgear	GET	Fetch CMDB network devices

8. Authentication

Authentication is handled by ServiceNow's built-in session management.

- Role-Based Access Control (RBAC) — three roles: requester, approver, engineer
- ACLs enforce read, create, and write permissions at the record level
- Group membership controls task assignment
- Admin override available for emergency changes
- All actions are logged in the audit trail

9. User Interface

UI Components:

1. Service Portal — Network Request catalog item
2. Catalog Item Form — 5 variables with dynamic UI Policies
3. Requested Item Form — full record with status tracking
4. Catalog Task Form — task details and work notes
5. Approvals List — My Approvals view
6. Reports — 3 reports (Bar, Pie, List)
7. Dashboard — 3 widgets with live metrics

Screenshots:

[Add screenshots of each UI component]

10. Testing

Testing Strategy:

1. Unit Testing — each feature tested individually
2. Integration Testing — flow and notification interactions
3. End-to-End Testing — full lifecycle from submission to closure
4. UAT — 25 test cases, all passed

Testing Tools:

- ServiceNow Flow Designer Execution Details
- System Log > Emails
- Task SLA records
- Reports and Dashboard

- Impersonation for role-based testing

11. Screenshots or Demo

Demo Video Link:

https://drive.google.com/drive/folders/1wRiiCeXa7NAmKIRUE-VMk6SOPfgyQrDv?usp=drive_link

12. Known Issues

1. Flow trigger initially did not fire — fixed by changing trigger table to sc_req_item
2. UI Policy not hiding mandatory fields — fixed execution order in client script
3. Duplicate approval notifications — deleted duplicate records
4. Group reference in global flow — created Network Approvers Global
5. Escalation flow blocking task creation — restructured flow to prevent blocking

All known issues have been resolved.

13. Future Enhancements

1. Multi-level approvals for high-risk requests
2. Integration with actual network devices for automatic configuration
3. Natural language intake using Virtual Agent
4. Mobile-friendly Service Portal
5. Advanced SLA analytics and dashboards
6. Bulk request submission
7. Knowledge base integration
8. Email preference center for users
9. Advanced reporting with Performance Analytics
10. Integration with change management for firewall changes